



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Multi AI Agents Workflow for Content Creation (Claude Edition)

TGS Ref No: TGS-2023036153

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	13 Sep 2026	Multi-agent content strategy courseware aligned to the current course page and TSC; GreenLeaf Café case; 12 labs.	Tertiary Infotech Academy
2.0	8 Oct 2026	Revamped in the house design of the Agentic AI Applications with Codex course. One central case — Horizon Wealth Planning, a Singapore financial-planning firm — across 14 labs, each in its own folder: Claude Cowork sub-agents (research, ideation, personas, cadence, skills, measurement), Codex (website, compliance subagent, approval gate, video), Claude Code subagents and agent teams (social, newsletter and video teams with LinkedIn, Facebook and YouTube publishing behind a human approval gate) and Hermes Agent bots on a Kanban board. Practice Exam slide (Claude Certified Associate — Foundations). WA (6 SAQ, K1-K6) and PP (4 tasks, A1-A7) rewritten for the v2.0 labs; assessment 4:00-6:00 PM Day 2 (WA 1 h, PP 1 h).	Dr. Alfred Ang
2.1	8 Oct 2026	Two Claude tools only: Claude (Chat and Cowork are one product) and Claude Code, which replaces Codex for the website, compliance subagent, approval gate and video. The website is published and shared as a Claude artifact instead of GitHub Pages. New slides: Claude plans and pricing; the agentic loop, why it matters and the loop in action; permission modes; publishing a Claude artifact. Topic 4 runs a scheduled weekly brief in Claude (Lab 12) and the playbook (Lab 13); Hermes Agent becomes an optional demo at the end (Lab 14). PP Task 4 updated to the Topic 4 labs.	Dr. Alfred Ang
3.0	8 Oct 2026	Two editions from one source — Claude and ChatGPT — each with its own deck, Lesson Plan, Learner Guide, labs and assessment. Rebuilt around an AI marketing team: a Marketing Team Lead and five specialists (Growth Strategist, Content Creator, Creative Designer, Website Designer, Growth Analyst), each with instructions, two skills and its own connectors. The team runs five campaigns: research to storyboard, a social week, newsletter and landing page, a YouTube explainer and an always-on weekly report. Claude edition: Claude Code subagents and agent teams; the website as a Claude artifact; optional Hermes Agent demo. 13 labs.	Dr. Alfred Ang
3.1	9 Oct 2026	Horizon's website is given, not built: Lab 1 now opens one public link (a shared Claude artifact) with a GitHub Pages backup for learners who cannot open artifacts or Sites, and adds the link to the team charter. The website links are printed and clickable on the scenario slide; the Practice Exam slide links to the exam. Lab 4 checks the public site; broken site URLs fixed.	Dr. Alfred Ang

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Course Overview

A two-day, hands-on WSQ course in content strategy with multi-agent AI workflows. Learners take one business — Horizon Wealth Planning, an independent financial-planning firm in Singapore with a website and no marketing team — and build it an AI marketing team: a Marketing Team Lead and five specialists (Growth Strategist, Content Creator, Creative Designer, Website Designer, Growth Analyst), each with instructions, skills and connectors. Day 1 forms the team and runs its first campaign; Day 2 runs a social week, a newsletter with a landing page, a YouTube explainer and an always-on weekly report, with a person approving every external piece. This Claude Edition uses Claude Code subagents and agent teams, Claude connectors and scheduled tasks, and publishes pages as Claude artifacts.

Course code TGS-2023036153 · 2 days · 16 hours including a 2-hour assessment · 135-slide deck v3.1 · 13 labs, each in its own folder. TSC: ICT-SNM-4004-1.1 Content Strategy.

Learning Outcomes

By the end of this course, participants will be able to:

- **LO1.** Conceptualize content ideas to meet marketing objectives and map out digital storyboards as part of a content strategy.
- **LO2.** Identify content requirements from customer preferences and determine the frequency of marketing content.
- **LO3.** Determine content types and styles and decide on modes and processes for distributing content.
- **LO4.** Develop guidelines for content strategy execution using appropriate delivery modes and responsible AI practices.

Daily Schedule

Both days run 9:00 AM – 6:00 PM with two 10-minute tea breaks and a 45-minute lunch. The Slides column gives the deck pages for each teaching block.

Day 1 — Form the Team

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:20 AM	20 min	Welcome, attendance, trainer, ground rules, learning outcomes, the Horizon scenario and your AI marketing team, lesson plan, assessment briefing	Presentation, discussion	1–19
9:20 AM–10:00 AM	40 min	Topic 1 — from one chatbot to a team of agents; the agentic loop; the tools and plans; a home for the team	Presentation, live demo	20–28
10:00 AM–10:30 AM	30 min	Lab 1 — Set up the studio and explore Horizon's website	Hands-on lab	29–34
10:30 AM–10:40 AM	10 min	Tea break	—	—
10:40 AM–10:55 AM	15 min	Agent instructions: job, inputs, outputs, limits	Presentation, live demo	35–36

10:55 AM–11:40 AM	45 min	Lab 2 — Form the team: instructions for each agent	Hands-on lab	37–41
11:40 AM–11:50 AM	10 min	Skills: twelve procedures, owned by role	Presentation	42–43
11:50 AM–12:40 PM	50 min	Lab 3 — Give each agent its skills	Hands-on lab	44–48
12:40 PM–1:25 PM	45 min	Lunch	—	—
1:25 PM–1:35 PM	10 min	Connectors and least privilege	Presentation	49–50
1:35 PM–2:20 PM	45 min	Lab 4 — Connect the Tools Each Agent Needs	Hands-on lab	51–55
2:20 PM–2:30 PM	10 min	Campaign 1: how the team works; the five-beat storyboard; regulated content	Presentation	56–58
2:30 PM–3:20 PM	50 min	Lab 5 — Campaign 1: research to storyboard, with subagents	Hands-on lab	59–63
3:20 PM–3:30 PM	10 min	Tea break	—	—
3:30 PM–3:35 PM	5 min	Topic 1 recap	Discussion	64
3:35 PM–3:45 PM	10 min	Topic 2 — evidence before personas; data agents count	Presentation	65–67
3:45 PM–4:25 PM	40 min	Lab 6 — Growth Analyst: personas from evidence	Hands-on lab	68–72
4:25 PM–4:35 PM	10 min	Finding the right frequency	Presentation	73
4:35 PM–5:10 PM	35 min	Lab 7 — Growth Analyst and Strategist: cadence and calendar	Hands-on lab	74–78
5:10 PM–6:00 PM	50 min	Day 1 review, Q&A and catch-up	Discussion	79

Day 2 — Run the Campaigns

Time	Duration	Topic / Activity	Method	Slides
9:00 AM–9:20 AM	20 min	Topic 3 — subagents vs agent teams; human in the loop, enforced; the publishing APIs	Presentation, live demo	80–87
9:20 AM–10:10 AM	50 min	Lab 8 — Campaign 2: a social media week	Hands-on lab	88–92
10:10 AM–10:20 AM	10 min	Tea break	—	—
10:20 AM–10:25 AM	5 min	Campaign 3: newsletter, landing page and consent	Presentation	93–94
10:25 AM–11:10 AM	45 min	Lab 9 — Campaign 3: newsletter and a landing page	Hands-on lab	95–99
11:10 AM–11:15 AM	5 min	Campaign 4: the video team	Presentation	100
11:15 AM–12:00 PM	45 min	Lab 10 — Campaign 4: the YouTube explainer	Hands-on lab	101–105
12:00 PM–12:05 PM	5 min	Topic 3 recap	Discussion	106
12:05 PM–12:50	45 min	Lunch	—	—

PM				
12:50 PM–1:05 PM	15 min	Topic 4 — always on: from skill to schedule	Presentation, live demo	107–109
1:05 PM–1:40 PM	35 min	Lab 11 — Campaign 5: the weekly growth report	Hands-on lab	110–113
1:40 PM–1:50 PM	10 min	Responsible AI for financial content; measuring what matters	Presentation	114–115
1:50 PM–2:30 PM	40 min	Lab 12 — Measure, govern and write the team playbook	Hands-on lab	116–120
2:30 PM–2:40 PM	10 min	Tea break	—	—
2:40 PM–2:50 PM	10 min	Optional demo — Hermes Agent	Presentation	121–122
2:50 PM–3:15 PM	25 min	Lab 13 — Optional Demo: the Team on Hermes Agent	Trainer demo	123–126
3:15 PM–4:00 PM	45 min	Course summary: the team, the campaigns, how each pattern is triggered; Q&A and assessment briefing	Discussion	127–128
4:00 PM–5:00 PM	60 min	Written Assessment (WA) — open book, 1 hour	Assessment	—
5:00 PM–6:00 PM	60 min	Practical Performance (PP) — 1 hour; TRAQOM, attendance, submit, sign the record	Assessment	129–135

Topic-by-topic Breakdown

Topic 1 — Multi-AI-Agent Content Ideation and Digital Storyboarding • Slides 20–64

- From one chatbot to a team of agents
- A home for the team; Horizon's public website
- Form the team: instructions, skills, connectors
- Campaign 1: research, ideas and a storyboard

Lab 1 — Set Up the Studio and Explore Horizon's Website • Slides 29–34

- **Surface:** Claude Code
- **Time:** 30 minutes (Day 1)
- **Features practised:** CLAUDE.md as the team charter • Horizon's public website
- **Output:** horizon-studio/ with CLAUDE.md, the shared data, a local copy of the site, and the public site link in the charter
- **Evidence:** Claude shows horizon-studio as the chosen folder. The public site opens on a phone; the calculator shows S\$693,138.
- **Lab folder:** labs_claude/lab-01-studio-and-website/

Lab 2 — Form the Team: Instructions for Each Agent • Slides 37–41

- **Surface:** Claude Code (subagent files)
- **Time:** 45 minutes (Day 1)

- **Features practised:** five subagent files in `.claude/agents/` · name, description, tools, model · the Lead is your session
- **Output:** `.claude/agents/`: growth-strategist, content-creator, creative-designer, website-designer, growth-analyst
- **Evidence:** Five files exist in `.claude/agents/`, one per specialist. You changed one instruction and can say why.
- **Lab folder:** `labs_claude/lab-02-agent-instructions/`

Lab 3 — Give Each Agent Its Skills · Slides 44–48

- **Surface:** Claude Code (`.claude/skills/`) → Claude
- **Time:** 50 minutes (Day 1)
- **Features practised:** twelve skills: two per specialist, two for the Lead · `SKILL.md` · the "Use when" trigger · test without naming
- **Output:** twelve skills in `.claude/skills/`, wired to the agents that own them
- **Evidence:** Twelve folders in `.claude/skills/`, each with a `SKILL.md`. Three skills also appear in Claude under Customize → Skills.
- **Lab folder:** `labs_claude/lab-03-agent-skills/`

Lab 4 — Connect the Tools Each Agent Needs · Slides 51–55

- **Surface:** `claude.ai` connectors → Claude Code (`/mcp`)
- **Time:** 45 minutes (Day 1)
- **Features practised:** Google Drive · Gmail (drafts) · Canva · Playwright MCP · least privilege per agent
- **Output:** every specialist tested on its own connector, with its limits written into its instructions
- **Evidence:** `/mcp` lists Google Drive, Gmail, Canva and Playwright. Each agent's instructions list only the tools its job needs.
- **Lab folder:** `labs_claude/lab-04-agent-connectors/`

Lab 5 — Campaign 1: Research to Storyboard, with Subagents · Slides 59–63

- **Surface:** Claude Code (subagents in parallel)
- **Time:** 50 minutes (Day 1)
- **Features practised:** the Lead delegates · strategist and analyst in parallel · nine scored ideas · a human concept checkpoint · a five-beat storyboard
- **Output:** `research/` (competitors, funnel), `strategy/ideas.md`, `strategy/storyboard.md` with three channel variants
- **Evidence:** The strategist and the analyst ran in parallel. The Lead's fact-check and fin-compliance report came back.
- **Lab folder:** `labs_claude/lab-05-campaign-1-research-to-storyboard/`

Topic 2 — Audience Research and Content Requirement Analysis · Slides 65–79

- The Growth Analyst: personas from evidence
- Content requirements every agent can follow
- How often to publish: value, fatigue and capacity
- The calendar the team will run

Lab 6 — Growth Analyst: Personas from Evidence • Slides 68–72

- **Surface:** Claude Code (analyst subagents)
- **Time:** 40 minutes (Day 1)
- **Features practised:** three analyst subagents, one per data file · counts, not impressions · confidence labels · content requirements
- **Output:** strategy/personas.md and strategy/content-spec.md
- **Evidence:** Three analyst subagents ran, one per file. content-spec.md has every field for every persona.
- **Lab folder:** labs_claude/lab-06-personas-from-evidence/

Lab 7 — Growth Analyst and Strategist: Cadence and Calendar • Slides 74–78

- **Surface:** Claude Code (subagents)
- **Time:** 35 minutes (Day 1)
- **Features practised:** marginal value vs fatigue · team capacity · review buffers · a four-week calendar
- **Output:** strategy/cadence.md and strategy/calendar.csv
- **Evidence:** cadence.md gives a frequency per channel with the evidence. No week is over 12 hours.
- **Lab folder:** labs_claude/lab-07-cadence-and-calendar/

Topic 3 — Multi-Channel Content Creation and Agent Workflow Coordination • Slides 80–106

- Subagents or an agent team?
- Human in the loop — enforced, not requested
- Campaigns 2–4: social week, newsletter + landing page, video
- From approval to LinkedIn, Facebook, email and YouTube

Lab 8 — Campaign 2: A Social Media Week • Slides 88–92

- **Surface:** Claude Code (agent team) → LinkedIn, Facebook
- **Time:** 50 minutes (Day 2)
- **Features practised:** the publisher kit and its hook · an agent team: creator, designer, Lead · teammates message each other · human approval → post
- **Output:** content/social/week-01/: five posts with images, reviewed, approved by you and posted (or dry-run)
- **Evidence:** The hook (or approve.mjs) stopped the agent from approving. Dry runs printed every request; nothing went live unless you said go.
- **Lab folder:** labs_claude/lab-08-campaign-2-social-week/

Lab 9 — Campaign 3: Newsletter and a Landing Page • Slides 95–99

- **Surface:** Claude Code (agent team) → Claude artifact, Gmail
- **Time:** 45 minutes (Day 2)
- **Features practised:** creator writes, website designer builds the lead-magnet page, analyst filters by consent, the Lead checks · drafts only
- **Output:** content/newsletter/2026-11/ (issue, HTML, text, recipients, excluded), web/checklist/ published, Gmail drafts

- **Evidence:** Three specialists worked on the campaign; the Lead checked all. Drafts only — nothing was sent to the list.
- **Lab folder:** labs_claude/lab-09-campaign-3-newsletter-and-landing-page/

Lab 10 — Campaign 4: The YouTube Explainer • Slides 101–105

- **Surface:** Claude Code (agent team, render, upload)
- **Time:** 45 minutes (Day 2)
- **Features practised:** strategist angle · creator script · designer storyboard, thumbnail and render · approve twice · private upload
- **Output:** content/video/ep01/: scenes.json, thumbnail, ep01.mp4, ep01.md, and a private YouTube upload (or dry run)
- **Evidence:** The strategist, creator and designer each delivered their part. The upload is private (or the dry run printed the request).
- **Lab folder:** labs_claude/lab-10-campaign-4-youtube-explainer/

Topic 4 — Content Distribution, Strategy Guidelines and Responsible AI Practices • Slides 107–135

- Campaign 5: always on, with a scheduled task
- Responsible AI for financial content
- Measure the patterns; write the team playbook
- Optional demo: the team on Hermes Agent

Lab 11 — Campaign 5: Always On — the Weekly Growth Report • Slides 110–113

- **Surface:** Claude (skill, scheduled task, Drive, Gmail)
- **Time:** 35 minutes (Day 2)
- **Features practised:** the Growth Analyst's campaign-report skill · do · verify · schedule · draft-only delivery
- **Output:** a scheduled task that drafts the weekly growth report every Monday at 8am, and one draft in Gmail
- **Evidence:** The campaign-report skill is available in Claude. The task is paused after class.
- **Lab folder:** labs_claude/lab-11-campaign-5-weekly-growth-report/

Lab 12 — Measure, Govern and Write the Team Playbook • Slides 116–120

- **Surface:** Claude Code (subagents)
- **Time:** 40 minutes (Day 2)
- **Features practised:** compare agent patterns · the roster: instructions, skills, connectors · approval matrix · responsible-AI controls
- **Output:** reports/agent-comparison.md and strategy/team-playbook.md
- **Evidence:** agent-comparison.md shows the one-agent baseline caught 0 defects. It has an incident procedure, and your own rule.
- **Lab folder:** labs_claude/lab-12-team-playbook/

Lab 13 — Optional Demo: the Team on Hermes Agent • Slides 123–126

- **Surface:** Hermes Agent (profiles, Kanban) — trainer demo
- **Time:** 25 minutes (Day 2)

- **Features practised:** open-source Hermes Agent · one profile per role · a Kanban pipeline · a task blocked for a person · the same skills
- **Output:** three Hermes profiles and a pipeline on the board, with the review task blocked for a person (demo)
- **Evidence:** hermes profile list shows the three profiles. Unblocking finished the pipeline.
- **Lab folder:** labs_claude/lab-13-optional-hermes-agent-demo/

Resources Required

- Trainer: the v3.1 Claude Edition slide deck, this Lesson Plan, the Learner Guide and the lab pack (labs_claude/, 13 folders, with reference solutions for Labs 1, 2, 3, 8 and 10).
- Each learner: a laptop with the Claude desktop app and the Claude Code CLI on a Pro plan or above (Claude Code and artifacts are not on the Free plan), signed in with /login.
- Node.js 22, Python 3 with Pillow, ffmpeg, and a personal Google account (Drive, Gmail). Canva account for the Canva connector.
- Optional demo (Lab 13): a spend-limited Anthropic or OpenRouter API key for Hermes Agent, held by the trainer.
- Optional, for live publishing: a LinkedIn developer app, a test Facebook Page and a Google Cloud project with the YouTube Data API. Every lab also works in dry-run mode without them.
- All fixtures are synthetic. Every learner email address resolves to the learner's own Gmail via plus-addressing.

Assessment

- Day 2, 4:00 – 6:00 PM: Written Assessment (WA, short-answer, 1 hour) from 4:00 PM, then Practical Performance (PP, 1 hour) from 5:00 PM.
- Open book: slides, Learner Guide and approved materials only.
- Funding criteria: minimum 75% attendance, and assessed as Competent in both the WA and the PP.
- Assessment flow: TRAQOM survey → assessment digital attendance → WA and PP → submit answers on the LMS → sign the Assessment Summary Record.
- Courseware and assessment are on the LMS: <https://lms-tms.tertiaryinfotech.com/>